Harsha Vardhan Yellela

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SUMMARY

AI Automation Engineer with hands-on experience building **multi-agent systems**, **MCP-based tool integrations**, and **LLM-powered automation pipelines** for production environments. Designed agentic workflows using **CrewAI, LangChain, and n8n** with **Amazon Bedrock** and **function calling**, achieving **70% reduction in manual processes**. Built **94 serverless functions** integrating **10+ third-party APIs** and **8-microservice MLOps platforms** with auto-retraining loops. Skilled in **RAG pipelines**, **structured output generation**, and **end-to-end automation** from data ingestion to deployment.

EXPERIENCE

Student Consultant – AI/LLM **Feb 2026 – Present**
Open Avenues (Build Fellowship) *Remote*

- Building a **GPT transformer from first principles** in PyTorch — developing hands-on understanding of attention mechanisms, tokenization strategies, and LLM training dynamics that directly inform production LLM integration and agentic AI work.
- Implementing and comparing tokenization strategies (char-level, GPT-2 BPE, cl100k, custom BPE via SentencePiece) to understand vocabulary-compression-training tradeoffs relevant to LLM prompt engineering and context management.
- Deploying fine-tuned model to **HuggingFace Spaces** with Gradio UI; building CLI tooling for training, evaluation, and interactive generation — full ownership from raw data to public demo.

Graduate Research Assistant – Agentic AI **Jan 2025 – Dec 2025**
Lawrence Technological University *Southfield, MI*

- Built and benchmarked **no-code (n8n)** vs. **coded multi-agent systems (CrewAI + LangChain)** with **MCP tool integration** for automated research workflows and intelligent decision-making.
- Deployed persistent **MCP agent services** on **AWS Fargate** and **Amazon EKS**, integrating **OpenSearch Serverless** for semantic search and **RAG-based automation**.
- Designed hybrid pipelines combining **Amazon Bedrock** foundation models with **function calling** and custom tool integrations, achieving **70% reduction in manual process time**.

Infor India Pvt. Ltd. **Apr 2022 – Dec 2023**
LN Technical Consultant *Hyderabad, India*

- Developed modular, production-ready **automation tools** for global clients (**Ferrari, Boeing, Triumph**) by extending **Infor LN ERP** workflows with custom business logic.
- Integrated **Infor ION process flows** with **AWS S3, Lambda, and API Gateway** for **asynchronous file transfer** and **event-driven automation**, reducing batch processing delays by **~40%**.
- **Containerized** business logic services using **Docker** and orchestrated with **Infor ION** for automated enterprise workflows across manufacturing and finance modules.

Junior Java Trainee (PEP Program) **Sep 2020 – Jul 2021**
EPAM Systems *Hyderabad, India*

- Completed structured **Java Core** training with exposure to **automated testing (JUnit)**, **CI/CD pipeline fundamentals**, and **Git-based version control** — establishing systematic software delivery practices.
- Operated in **Agile/Scrum** sprints applying clean-code and structured delivery principles; foundations directly carried into later automation engineering and agentic AI workflow development.

TECHNICAL SKILLS

- **Languages:** Python, Go, TypeScript, JavaScript, SQL, Bash
- **AI Agents & Orchestration:** CrewAI, LangChain, LangGraph, n8n | MCP (Model Context Protocol) | Function Calling, Structured Output
- **LLMs & Platforms:** GPT-4, Gemini, Claude API, Llama, Qwen | Amazon Bedrock | QLoRA Fine-tuning (PEFT) | Ollama, vLLM
- **RAG & Vector DBs:** Qdrant, pgvector, OpenSearch | Embeddings, BM25, Cross-Encoder Reranking | Semantic Search Pipelines
- **Cloud & DevOps:** AWS (Lambda, Fargate, EKS, SageMaker, API Gateway) | Docker, Kubernetes | Terraform, GitHub Actions
- **Backend & Integration:** FastAPI, Flask, Gin (Go) | Stripe, Twilio, DocuSign, QuickBooks | Event-Driven Architectures

PROJECTS

AgenticAI – Multi-Agent Newsletter Automation **Apr 2025**
CrewAI, LangChain, n8n, FastAPI, AWS Lambda | GitHub

- Designed and deployed **newsletter automation** using **n8n (no-code)** and **CrewAI + LangChain (code-first)** with **structured output parsing**, benchmarking both across latency, reliability, and developer speed.
- Built **multi-agent coordination** with **function calling**, custom **FastAPI** backends, and deployed on **AWS Lambda** for scalable, event-driven execution.
- Achieved **70% reduction in manual workload** and provided actionable insights into visual vs. programmatic agent orchestration tradeoffs.

CRE Research Agent – AI-Powered Literature Review **Nov 2024 – Present**
LangGraph, FastAPI, vLLM, pgvector, Qdrant, Elasticsearch, Redis

- Built automated **literature review system** using **multi-step RAG pipeline**: query expansion → retrieval → reranking → summarization, discovering **~20 relevant papers** per query.
- Implemented **LangGraph-style state machine** orchestration with **dual vector databases** (Qdrant + pgvector), **BM25 sparse retrieval**, and **BGE cross-encoder reranking**.
- Integrated **Semantic Scholar, ArXiv, and Crossref APIs** for automated paper discovery with inline citation tracking and bibliography generation.

N8N Workflow Automation – Self-Hosted AI Platform **Sep 2024**
n8n, OpenAI, SerpAPI, Tavily AI, Ollama, Qdrant, Docker | *GitHub*

- Built **newsletter automation workflow** with **AI agents** integrating **SerpAPI** for trending topic discovery and **Tavily AI** for research.
- Deployed **self-hosted AI starter kit** with **Docker Compose**, configuring **Qdrant vector database** for semantic search and **Ollama (Llama 3.2)** for local LLM inference.
- Designed **end-to-end automated pipelines** from topic discovery to structured content generation with zero manual intervention.

Lambda Microservices – Enterprise Serverless Integration Platform **2024**
AWS Lambda, DynamoDB, API Gateway, Terraform, CloudFormation | Stripe, Twilio, DocuSign, QuickBooks

- Developed **94 AWS Lambda functions** for a complete SaaS platform with **automated deployment pipelines** using Git change detection.
- Integrated **10+ third-party services**: **Stripe Connect** (vendor onboarding/payments), **DocuSign** (e-signature workflows), **Twilio** (communications), **QuickBooks** (accounting sync), **EagleView** (aerial imagery).
- Built **event-driven architectures** with **API Gateway** routing and **Terraform/CloudFormation IaC** for repeatable infrastructure provisioning.

ML Sentiment Feedback Loop – Automated MLOps Platform **Nov 2024 – Present**
AWS (ECS Fargate, SageMaker, S3), Terraform, GitHub Actions, Docker | *GitHub*

- Built **8-microservice MLOps architecture** with **automated feedback loops**: Inference → Feedback → Evaluation → Retraining → Deployment, requiring **zero manual intervention**.
- Implemented **auto-retraining triggers** with **SageMaker**, automated **model versioning/registry**, and **SNS-based notifications** for pipeline events.
- Configured **GitHub Actions CI/CD** with **Terraform IaC** for fully automated infrastructure provisioning and container deployments to **ECS Fargate**.

AI Resume Builder (Resumade.in) – Multi-LLM Automation **Jun 2025 – Present**
FastAPI, AWS Lambda, DynamoDB, GPT-4, Gemini, DeepSeek, GitHub Actions

- Architected **serverless automation pipeline** using **FastAPI + AWS Lambda** with **multi-LLM orchestration** (GPT-4, Gemini, DeepSeek) and **structured JSON output** for intelligent resume generation.
- Built **real-time web scraping + AI optimization pipelines** using **BeautifulSoup** and **LinkedIn parsers**, reducing manual resume tailoring effort by **80%**.
- Deployed with **95%+ uptime**, **sub-2s PDF/Word generation**, and **GitHub Actions CI/CD** for automated testing and deployment.

LuffyGPT – GPT Transformer Language Model from Scratch **Jan 2026 – Mar 2026**
Python, PyTorch, SentencePiece, tiktoken, Gradio, HuggingFace Hub | *GitHub* | *Demo*

- Implemented a **GPT decoder-only transformer from scratch** in PyTorch (10.81M params) — multi-head causal self-attention, positional embeddings, FFN with 4× expansion, residual connections, pre-LayerNorm, dropout; followed the *Attention is All You Need* architecture.
- Trained a **custom BPE tokenizer** using SentencePiece on a ~3.2M character corpus (vocab 2000, 3.11× compression); benchmarked char-level, GPT-2 BPE, cl100k, and o200k strategies; trained on **dual NVIDIA RTX 3090 GPUs** (DataParallel), reaching **val loss ~1.18** at 25K steps.
- Deployed as a **Gradio app on HuggingFace Spaces** with model weights hosted on HuggingFace Hub; built a CLI tool supporting `--train`, `--eval`, `--interactive`, and `--custom-bpe` modes.

EDUCATION

Lawrence Technological University **Dec 2025**
*Master of Science in Computer Science · GPA: **3.6/4.0*** *Southfield, MI*

- **Relevant Coursework: Agentic AI Research**, **Machine Learning**, **Natural Language Processing**, **Artificial Intelligence**, **Intelligent Robotics (ROS)**

Geethanjali College of Engineering & Technology **Aug 2022**
*Bachelor of Technology in Computer Science & Engineering · GPA: **7.5/10 (~3.0/4.0)*** *Hyderabad, India*

- **Relevant Coursework: Deep Learning & Python**, **Machine Learning Foundations**, **Software Engineering**, **Internet of Things**

ACHIEVEMENTS

- **Selected for Amazon Nova AI Challenge: Trusted AI Track** (2025)
- **Built production SaaS platform** with 94 Lambda functions and 10+ third-party integrations serving real customers (2024)
- **Participated in RSNA Pneumonia Detection Challenge**; ranked in upper quartile with VGG19 transfer learning model (2024)
- **Gold Medalist in Indian National Mathematical Olympiad (INMO)** (2012)